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Optical system and image pickup apparatus having the same including five lens units of $+ - + + -$ or $+ - - + -$ refractive powers

Abstract

An optical system includes, in order from an object side to an image side, a first lens unit having positive refractive power, a second lens unit having negative refractive power, a third lens unit having positive or negative refractive power, a fourth lens unit having positive refractive power, and a fifth lens unit having negative refractive power. During focusing from infinity to the shortest distance, the second lens unit and the fourth lens unit move, the first lens unit, the third lens unit, and the fifth lens unit do not move. A predetermined condition is satisfied.

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Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
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2020060661	12/2019	JP	N/A

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Background/Summary

BACKGROUND

Technical Field

(1) One of the aspects of the disclosure relates to an optical system, which is suitable for a digital video camera, a digital still camera, a broadcasting camera, a film-based camera, a surveillance camera, an in-vehicle camera, and the like.

Description of the Related Art

(2) There has recently been demanded an image pickup apparatus that has a long focal length, a large aperture ratio, a small size, and high optical performance, and can achieve high-speed autofocus (AF). In order to realize the high-speed AF with the small size and high optical performance, an inner focus optical system is known in which part of the lenses is moved during focusing (see Japanese Patent Laid-Open Nos. 2014-142601 and 2020-060661).

(3) However, optical systems having the long focal length and large aperture ratio tend to have difficulty in correcting various aberrations. In particular, if the weight of the focus lens unit is reduced for faster AF, it becomes difficult to suppress aberration fluctuations during focusing from infinity to the shortest distance.

SUMMARY

(4) The disclosure provides an optical system that has a long focal length, a large aperture ratio, a small size, and high optical performance, and can realize high-speed autofocus, and an image pickup apparatus having the same.

(5) An optical system according to one aspect of the disclosure includes, in order from an object side to an image side, a first lens unit having positive refractive power, a second lens unit having negative refractive power, a third lens unit having positive or negative refractive power, a fourth lens unit having positive refractive power, and a fifth lens unit having negative refractive power.

During focusing from infinity to the shortest distance, the second lens unit and the fourth lens unit move, the first lens unit, the third lens unit, and the fifth lens unit do not move. The following inequalities are satisfied:

$$0.10 < sk/f < 0.25$$

$$0.010 < f/|f_3| < 0.500$$

$$-3.50 < f/f_5 < -0.80$$

$$-1.6 < f/f_2 < -0.3$$

where sk is a distance on an optical axis from a lens surface closest to an image plane of the optical system to the image plane, f is a focal length of the optical system in an in-focus state at infinity, and f_3 is a focal length of the third lens unit, f_5 is a focal length of the fifth lens unit, and f_2 is a focal length of the second lens unit. An image pickup apparatus according to another aspect of the disclosure includes the above optical system.

(6) Further features of the disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a sectional view of an optical system according to Example 1 in an in-focus state at infinity (on an infinity object).

(2) FIGS. 2A and 2B are longitudinal aberration diagrams of the optical system according to Example 1 in the in-focus state at infinity and an in-focus state at a short distance (on a short-distance object) (−700 mm).

(3) FIG. 3 is a sectional view of an optical system according to Example 2 in an in-focus state at infinity.

(4) FIGS. 4A and 4B are longitudinal aberration diagrams of the optical system according to Example 2 in the in-focus state at infinity and an in-focus state at a short distance (−700 mm).

(5) FIG. 5 is a sectional view of an optical system according to Example 3 in an in-focus state at infinity.

(6) FIGS. 6A and 6B are longitudinal aberration diagrams of the optical system according to Example 3 in the in-focus state at infinity and an in-focus state at a short distance (−700 mm).

(7) FIG. 7 is a sectional view of an optical system according to Example 4 in an in-focus state at infinity.

(8) FIGS. 8A and 8B are longitudinal aberration diagrams of the optical system according to Example 4 in the in-focus state at infinity and an in-focus state at a short distance (−1200 mm).

(9) FIG. 9 is a sectional view of an optical system according to Example 5 in an in-focus state at infinity.

(10) FIGS. 10A and 10B are longitudinal aberration diagrams of the optical system according to Example 5 in the in-focus state at infinity and an in-focus state at a short distance (−700 mm).

(11) FIG. 11 is a schematic diagram of an image pickup apparatus.

DESCRIPTION OF THE EMBODIMENTS

(12) Referring now to the accompanying drawings, a detailed description will be given of embodiments according to the disclosure. Corresponding elements in respective figures will be designated by the same reference numerals, and a duplicate description thereof will be omitted.

(13) FIGS. 1, 3, 5, 7, and 9 are sectional views of optical systems L0 according to Examples 1 to 5, respectively, in an in-focus state at infinity. The optical system L0 according to each example is an optical system for an image pickup apparatus such as a digital video camera, a digital still camera, a broadcasting camera, a film-based camera, a surveillance camera, and an in-vehicle camera.

(14) In each sectional view, a left side is an object side and a right side is an image side. The optical

system **L0** according to each example includes a plurality of lens units. In the specification of this application, a lens unit is a group of lenses that are integrally moved or fixed during focusing. That is, in the optical system **L0** according to each example, a distance between adjacent lens units changes during focusing. The lens unit includes one or more lenses.

(15) The optical system **L0** according to each example includes, in order from the object side to the image side, a first lens unit **L1** having positive refractive power, a second lens unit **L2** having negative refractive power, a third lens unit **L3** having positive or negative refractive power, a fourth lens unit **L4** having positive refractive power, and a fifth lens unit **L5** having negative refractive power.

(16) **SP** denotes an aperture stop (diaphragm). **IMG** is an image plane. In a case where the optical system **L0** according to each example is used as an imaging optical system for a digital still camera or a digital video camera, an imaging plane of a solid-state image sensor (photoelectric conversion element) such as a CCD sensor or a CMOS sensor is placed on the image plane **IMG**. In a case where the optical system **L0** according to each example is used as an imaging optical system for a film-based camera, a photosensitive surface corresponding to the film plane is placed on the image plane **IMG**.

(17) An arrow illustrated in each sectional view represents a moving direction of a lens unit during focusing from infinity to the shortest distance (or the closest end). In the optical system **L0** according to each example, during focusing from infinity to the shortest distance, the second lens unit **L2** and the fourth lens unit **L4** move, and the first lens unit **L1**, the third lens unit **L3**, and the fifth lens unit **L5** do not move.

(18) **FIGS. 2A, 2B, 4A, 4B, 6A, 6B, 8A, 8B, 10A, and 10B** are aberration diagrams of the optical systems **L0** according to Examples 1 to 5, respectively. In each aberration diagram, **FIGS. 2A, 4A, 6A, 8A, and 10A** are longitudinal aberration diagrams of the optical systems **L0** according to Examples 1 to 5, respectively, in an in-focus state at infinity, and **FIGS. 2B, 4B, 6B, 8B, and 10B** are longitudinal aberration diagrams of the optical systems **L0** according to Examples 1 to 5, respectively, in an in-focus state at a short distance.

(19) In a spherical aberration diagram, **Fno** denotes an F-number, which indicates spherical aberration amounts for the d-line (wavelength 587.6 nm) and the g-line (wavelength 435.8 nm). In an astigmatism diagram, ΔS denotes an astigmatism amount on a sagittal image plane, and ΔM denotes an astigmatism amount on a meridional image plane. A distortion diagram illustrates a distortion amount for the d-line. A chromatic aberration diagram illustrates a chromatic aberration amount for the g-line. ω is an imaging half angle of view (degrees).

(20) A description will now be given of a characteristic configuration of the optical system **L0** according to each example.

(21) In order to achieve high optical performance, a long focal length, and a large aperture ratio with a small size and to achieve high-speed AF, the arrangement of the lens units included in the optical system **L0** and the configuration and arrangement of the focus lens unit that moves during focusing are important. The optical system **L0** according to each example includes a plurality of lens units and has a configuration in which part of the lens units are moved during focusing, thereby realizing aberration correction and weight reduction of the focus lens unit. The first lens unit **L1** having positive refractive power disposed closest to the object can reduce a diameter of an on-axis light beam (luminous flux) incident on the second lens unit **L2** and the fourth lens unit **L4**, which form the focus lens unit, and easily reduce the size and weight of the focus lens unit. The third lens unit **L3** disposed between the second lens unit **L2** and the fourth lens unit **L4**, which form the focus lens unit, can easily correct spherical aberration and longitudinal chromatic aberration. The fifth lens unit **L5** having negative refractive power disposed on the image side of the second lens unit **L2** and the fourth lens unit **L4**, which form the focus lens unit, can easily reduce the overall length of the optical system **L0** and suppress the Petzval sum.

(22) The optical system **L0** according to each example satisfies the following inequality (1) to (4):

$$0.10 < sk/f < 0.25 \quad (1)$$

$$0.010 < f/|f_3| < 0.500 \quad (2)$$

$$-3.50 < f/f_5 < -0.80 \quad (3)$$

$$-1.6 < f/f_2 < -0.3 \quad (4)$$

where sk is a distance on the optical axis from a lens surface closest to the image plane of the optical system **L0** to the image plane (referred to as back focus hereinafter). f is a focal length of the optical system **L0** in the in-focus state at infinity. f_3 is a focal length of the third lens unit **L3**. f_5 is a focal length of the fifth lens unit **L5**. f_2 is a focal length of the second lens unit **L2**.

(23) Inequality (1) defines a ratio between the back focus and the focal length of the optical system **L0**. In a case where the back focal length becomes so short that the value is lower than the lower limit of inequality (1), the layout near the image sensor becomes difficult and the configuration becomes complicated. In a case where the back focus becomes so long that the value is higher than the upper limit of inequality (1), the optical system **L0** becomes large.

(24) Inequality (2) defines a ratio between the focal length of the third lens unit **L3** and the focal length of the optical system **L0**. In a case where the refractive power of the third lens unit **L3** becomes so small that the value is lower than the lower limit of inequality (2), it becomes difficult to reduce the diameter of the on-axis light beam incident on the fourth lens unit **L4**, and the fourth lens unit **L4** becomes large. In a case where the refractive power of the third lens unit **L3** becomes so large that the value is higher than the upper limit of inequality (2), correction of spherical aberration becomes difficult.

(25) Inequality (3) defines a ratio between the focal length of the fifth lens unit **L5** and the focal length of the optical system **L0**. In a case where the refractive power of the fifth lens unit **L5** becomes so small that the value is lower than the lower limit of inequality (3), correction of distortion becomes difficult. In a case where the refractive power of the fifth lens unit **L5** becomes so large that the value is higher than the upper limit of inequality (3), it becomes difficult to shorten the overall length of the optical system **L0**, and the optical system **L0** becomes large.

(26) Inequality (4) defines a ratio between the focal length of the second lens unit **L2** and the focal length of the optical system **L0**. In a case where the refractive power of the second lens unit **L2** becomes so small that the value is lower than the lower limit of inequality (4), it becomes difficult to correct the spherical aberration generated in the second lens unit **L2**, and in particular, it becomes difficult to suppress the fluctuation of the spherical aberration during focusing. In a case where the refractive power of the second lens unit **L2** becomes so large that the value is higher than the upper limit of inequality (4), the moving amount of the second lens unit **L2** increases during focusing and the optical system **L0** becomes large.

(27) Inequalities (1) to (4) may be replaced with inequalities (1a) to (4a) below.

$$0.12 < sk/f < 0.22 \quad (1a)$$

$$0.011 < f/|f_3| < 0.450 \quad (2a)$$

$$-2.50 < f/f_5 < -1.20 \quad (3a)$$

$$-1.6 < f/f_2 < -0.5 \quad (4a)$$

(28) Inequalities (1) to (4) may be replaced with inequalities (1b) to (4b) below.

$$0.15 < sk/f < 0.20 \quad (1b)$$

$$0.012 < f/|f_3| < 0.420 \quad (2b)$$

$$-2.40 < f/f_5 < -1.50 \quad (3b)$$

$$-1.6 < f/f_2 < -0.6 \quad (4b)$$

(29) A description will now be given of configurations that may be satisfied by the optical system **L0** according to each example.

(30) The second lens unit **L2** may include, in order from the object side to the image side, a positive lens **L21** and a negative lens **L22**. In the second lens unit **L2**, the positive lens **L21** disposed on the object side can suppress the lens diameter of the succeeding negative lens **L22**, and is advantageous to miniaturization. The second lens unit **L2** may include two lenses, that is, the

positive lens L21 and the negative lens L22, so that the weight can be easily reduced.

(31) A biconcave air lens may be formed between the positive lens L21 and the negative lens L22. This configuration can easily suppress fluctuations in spherical aberration occurring in the second lens unit L2, and easily achieve high image quality over the entire focus range.

(32) The fourth lens unit L4 may include, in order from the object side to the image side, a positive lens L41, a negative lens L42, and a positive lens L43, and the positive lens L41 and the negative lens L42 may be formed as a cemented lens. The cemented lens facilitates correction of longitudinal chromatic aberration. The fourth lens unit L4 including two positive lenses can disperse the refractive power, thereby facilitating correction of aberrations over the entire focal range.

(33) The third lens unit L3 may include, in order from the object side to the image side, a negative lens L31 and a positive lens L32, and the negative lens L31 and the positive lens L32 may be formed as a cemented lens. The cemented lens facilitates suppression of the Petzval sum and correction of longitudinal chromatic aberration.

(34) The third lens unit L3 may include the aperture stop SP. The third lens unit L3 including the aperture stop SP improves the symmetry of the refractive power arrangement before and after the aperture stop SP and facilitates corrections of distortion and coma.

(35) The positive lens L11 may be disposed closest to the object side. This configuration facilitates reducing the overall length of the optical system L0 and the size and weight of the focus lens unit.

(36) The first lens unit L1 may include two or more positive lenses and one or more negative lenses. This configuration facilitates shortening the overall length of the optical system L0 and suppressing longitudinal and lateral chromatic aberrations.

(37) The fifth lens unit L5 may include a positive lens and a negative lens. Since the fifth lens unit L5 is the lens unit closest to the image plane, it is effective in correcting the Petzval sum, and since the fifth lens unit L5 includes a positive lens and a negative lens, curvature of field can be easily corrected.

(38) A description will now be given of conditions that the optical system according to each example may satisfy. The optical system according to each example may satisfy one or more of the following inequalities (5) to (11).

$$0.4 < f/f_1 < 2.5 \quad (5)$$

$$1.0 < f/f_4 < 3.5 \quad (6)$$

$$0.7 < L/f < 2.0 \quad (7)$$

$$1.4 < b_2 < 4.0 \quad (8)$$

$$0.1 < b_4 < 0.8 \quad (9)$$

$$0.01 < M_2/f < 0.30 \quad (10)$$

$$-0.20 < M_4/f < -0.01 \quad (11)$$

(39) Here, f_1 is a focal length of the first lens unit L1. f_4 is a focal length of the fourth lens unit L4. L is a distance on the optical axis from a lens surface closest to the object of the optical system L0 to the image plane (referred to as an overall optical length (overall lens length) hereinafter). b_2 is an imaging lateral magnification of the second lens unit L2 in the in-focus state at infinity. b_4 is an imaging lateral magnification of the fourth lens unit L4 in the in-focus state at infinity. M_2 is a moving amount of the second lens unit L2 during focusing from infinity to the shortest distance, where a direction moving toward the image side is set positive. M_4 is a moving amount of the fourth lens unit during focusing from infinity to the shortest distance, where a direction moving toward the image side is set positive.

(40) Inequality (5) defines a ratio between the focal length of the first lens unit L1 and the focal length of the optical system L0. In a case where the refractive power of the first lens unit L1 becomes so small that the value is lower than the lower limit of inequality (5), it becomes difficult to shorten the overall length of the optical system L0, and the optical system L0 becomes large. In a case where the refractive power of the first lens unit L1 becomes so large that the value is higher

than the upper limit of inequality (5), the aberrations generated in the first lens unit L1 increase, and it becomes difficult to correct spherical aberration and lateral chromatic aberration.

(41) Inequality (6) defines a ratio between the focal length of the fourth lens unit L4 and the focal length of the optical system L0. In a case where the refractive power of the fourth lens unit L4 becomes so small that the value is lower than the lower limit of inequality (6), a moving amount of the fourth lens unit L4 during focusing becomes large and the optical system L0 becomes large. In a case where the refractive power of the second lens unit L2 becomes so large that the value is higher than the upper limit of inequality (6), it becomes difficult to correct spherical aberration generated in the fourth lens unit L4, and in particular, it becomes difficult to suppress the fluctuation of the spherical aberration during focusing.

(42) Inequality (7) defines a ratio between the overall optical length and the focal length of the optical system L0. In a case where the overall optical length becomes so small that the value is lower than the lower limit of inequality (7), it becomes difficult to correct aberrations, especially lateral chromatic aberration. In a case where the overall optical length becomes so large that the value is higher than the upper limit of inequality (7), the diameter of the front lens increases and the optical system L0 becomes large.

(43) Inequality (8) defines the imaging lateral magnification of in the in-focus state at infinity of the second lens unit L2. In a case where the imaging lateral magnification of the second lens unit L2 becomes so small that the value is lower than the lower limit of inequality (8), the moving amount of the second lens unit L2 for focusing becomes large and the optical system L0 becomes large. In a case where the lateral imaging magnification of the second lens unit L2 becomes so large that the value is higher than the upper limit of inequality (8), the heights of light rays incident on the succeeding lens units increase and the optical system L0 becomes large.

(44) Inequality (9) defines the imaging lateral magnification of the fourth lens unit L4 in the in-focus state at infinity. In a case where the imaging lateral magnification of the fourth lens unit L4 becomes so small that the value is lower than the lower limit of inequality (9), the incident angle of the on-axis light beam on the fifth lens unit L5 increases, and it becomes difficult to correct spherical aberration and coma. In a case where the imaging lateral magnification of the fourth lens unit L4 becomes so large that the value is higher than the upper limit of inequality (9), the moving amount of the fourth lens unit L4 or focusing increases, and the optical system L0 becomes large.

(45) Inequality (10) defines a ratio between the moving amount of the second lens unit L2 and the focal length of the optical system L0 during focusing from infinity to the shortest distance. In a case where the moving amount of the second lens unit L2 becomes so small that the value is lower than the lower limit of inequality (10), imaging in a range including the shortest distance becomes difficult. In a case where the moving amount of the second lens unit L2 becomes so large that the value is higher than the upper limit of inequality (10), the optical system L0 becomes large.

(46) Inequality (11) defines a ratio between the moving amount of the fourth lens unit L4 and the focal length of the optical system L0 during focusing from infinity to the shortest distance. In a case where the moving amount of the fourth lens unit L4 becomes so small that the value is lower than the lower limit of inequality (11), that is, in a case where the absolute value of the moving amount of the fourth lens unit L4 becomes large, the optical system L0 becomes large. In a case where the moving amount of the fourth lens unit L4 becomes so large that the value is higher than the upper limit of inequality (11), that is, in a case where the absolute value of the moving amount of the fourth lens unit L4 becomes small, imaging in a range including the shortest distance becomes difficult.

(47) Inequalities (5) to (11) may be replaced with inequalities (5a) to (11a) below.

$$0.5 < f/f_1 < 2.0 \quad (5a)$$

$$1.5 < f/f_4 < 3.0 \quad (6a)$$

$$0.8 < L/f < 1.8 \quad (7a)$$

$$1.6 < b_2 < 3.3 \quad (8a)$$

$$0.2 < b_4 < 0.7 \quad (9a)$$

$$0.02 < M_2/f < 0.15 \quad (10a)$$

$$-0.10 < M_4/f < -0.03 \quad (11a)$$

(48) Inequalities (5) to (11) may be replaced with inequalities (5b) to (11b) below.

$$0.6 < f/f_1 < 1.7 \quad (5b)$$

$$1.7 < f/f_4 < 2.4 \quad (6b)$$

$$0.9 < L/f < 1.7 \quad (7b)$$

$$1.7 < b_2 < 3.0 \quad (8b)$$

$$0.25 < b_4 < 0.6 \quad (9b)$$

$$0.03 < M_2/f < 0.13 \quad (10b)$$

$$-0.08 < M_4/f < -0.05 \quad (11b)$$

(49) A detailed description will be given of the optical system L0 according to each example.

(50) Each of the optical systems L0 according to Examples 1 to 4 includes, in order from the object side to the image side, a first lens unit L1 having positive refractive power, a second lens unit L2 having negative refractive power, a third lens unit L3 having positive refractive power, a fourth lens unit L4 having positive refractive power, and a fifth lens unit L5 having negative refractive power.

(51) The optical system L0 according to Example 5 includes, in order from the object side to the image side, a first lens unit L1 having positive refractive power, a second lens unit L2 having negative refractive power, a third lens unit L3 having negative refractive power, a fourth lens unit L4 having positive refractive power, and a fifth lens unit L5 having negative refractive power.

(52) In the optical system L0 according to each example, during focusing from infinity to the shortest distance, the second lens unit L2 and the fourth lens unit L4 move, and the first lens unit L1, the third lens unit L3, and the fifth lens unit L5 do not move.

(53) In the optical systems L0 according to Examples 1, 2, and 5, the first lens unit L1 includes, in order from the object side to the image side, a positive lens L11, a positive lens L12, a positive lens L13, and a negative lens L14. The positive lens L13 and the negative lens L14 are formed as a cemented lens.

(54) In the optical system L0 according to Example 3, the first lens unit L1 includes, in order from the object side to the image side, a positive lens L11, a positive lens L12, a positive lens L13, a negative lens L14, and a positive lens L15. The positive lens L13 and the negative lens L14 are formed as a cemented lens.

(55) In the optical system L0 according to Example 4, the first lens unit L1 includes, in order from the object side to the image side, a positive lens L11, a positive lens L12, a positive lens L13, a positive lens L14, a negative lens L15, and a positive lens L16. The positive lens L14 and the negative lens L15 are formed as a cemented lens.

(56) In the optical system L0 according to each example, the second lens unit L2 includes, in order from the object side to the image side, a positive lens L21 and a negative lens L22.

(57) In the optical system L0 according to each example, the third lens unit L3 includes, in order from the object side to the image side, a negative lens L31 and a positive lens L32. The negative lens L31 and the positive lens L32 are formed as a cemented lens.

(58) In the optical system L0 according to each example, the third lens unit L3 includes an aperture stop SP.

(59) In the optical system L0 according to each example, the fourth lens unit L4 includes, in order from the object side to the image side, a positive lens L41, a negative lens L42, and a positive lens L43. The positive lens L41 and the negative lens L42 are formed as a cemented lens.

(60) In the optical systems L0 according to Examples 1 to 3, the fifth lens unit L5 includes, in order from the object side to the image side, a positive lens L51, a negative lens L52, and a negative lens L53. The positive lens L51 and the negative lens L52 are formed as a cemented lens.

(61) In the optical system L0 according to Example 4, the fifth lens unit L5 includes, in order from the object side to the image side, a positive lens L51, a negative lens L52, and a positive lens L53.

The positive lens L51 and the negative lens L52 are formed as a cemented lens.

(62) In the optical system L0 according to Example 5, the fifth lens unit L5 includes, in order from the object side to the image side, a positive lens L51, a negative lens L52, a negative lens L53, and a positive lens L54. The positive lens L51 and the negative lens L52 are formed as a cemented lens.

(63) Numerical examples 1 to 5 corresponding to Examples 1 to 5 will be illustrated below.

(64) In surface data in each numerical example, r denotes a radius of curvature of each optical surface, and d (mm) denotes an on-axis distance (or a distance on the optical axis) between an m-th surface and an (m+1)-th surface, where m is a surface number counted from the light incident side. nd denotes a refractive index for the d-line of each optical element, and vd denotes an Abbe number of the optical element. The Abbe number vd of a certain material is expressed as follows:

$$vd=(Nd-1)/(NF-NC)$$

where Nd, NF, and NC are refractive indexes with respect to the d-line (587.6 nm), the F-line (486.1 nm), and the C-line (656.3 nm) in the Fraunhofer line, respectively.

(65) A location where the on-axis distance d is variable illustrates the in-focus state at infinity and the in-focus state at a short distance. A numerical value of an object distance in the in-focus state at the short distance is given in parenthesis. The object distance is a distance from the image plane to the object position.

(66) In each numerical example, each of d, focal length (mm), F-number, and half angle of view (degrees) has a value in a case where the optical system L0 according to each example is in the in-focus state on an object at infinity (infinity object). A “back focus” is a distance on the optical axis from the final lens surface (the lens surface closest to the image plane) to a paraxial image plane in terms of air equivalent length. An “overall lens length” is a length obtained by adding the back focus to a distance on the optical axis from the frontmost surface (lens surface closest to the object) to the final surface of the zoom lens.

(67) In a case where the optical surface is aspherical, an asterisk * is attached to the right side of the surface number. The aspherical shape is expressed as follows:

$$X=(h.\text{sup.}2/R)/[1+\{1-(1+K)$$

$$(h/R).\text{sup.}2\}.\text{sup.}1/2]+A4\times h.\text{sup.}4+A6\times h.\text{sup.}6+A8\times h.\text{sup.}8+A10\times h.\text{sup.}10+A12\times h.\text{sup.}12$$

where X is a displacement amount from a surface vertex in the optical axis direction, h is a height from the optical axis in a direction orthogonal to the optical axis, R is a paraxial radius of curvature, K is a conical constant, A4, A6, A8, A10, and A12 are aspherical coefficients of respective orders. “e±XX” in each aspherical coefficient means “×10.^{sup.}±XX.”

Numerical Example 1

(68) TABLE-US-00001 UNIT: mm Surface Data

Surface No.	r	d	nd	vd
1	77.785	4.68	1.76385	
48.5	2	172.845	0.20	3
63.371	5.95	1.49700	81.5	4
170.333	0.20	5	54.700	13.38
1.49700	81.5	6	-89.309	1.60
1.83400	37.2	7	115.572 (Variable)	8
686.201	3.52	1.94594	18.0	9
-143.362	0.20	10	457.173	1.40
1.67300	38.3	11	36.221 (Variable)	12(SP)
∞	4.29	13	-68.576	1.40
1.72825	28.5	14	34.701	9.22
1.87070	40.7	15	-101.270 (Variable)	16
40.392	4.86	1.72916	54.7	17
192.612	1.40	1.67270	32.1	18
76.112	2.50	19*	206.498	5.26
1.85135	40.1	20*	-91.873	
(Variable)	21	75.769	4.06	1.92286
20.9	22	-274.316	1.20	1.77047
29.7	23	29.541	8.48	24*
-73.133	3.03	1.58313	59.4	25*
-253.666	16.31	Image Plane ∞ Aspheric Data		
19th Surface	K = 0.00000e+00			
A 4 = -9.07618e-06	A 6 = -6.11165e-09	A 8 = -1.04661e-11	20th Surface	
K = 0.00000e+00	A 4 = -5.22186e-06	A 6 = -4.28155e-09	A 8 = -8.68689e-12	24th Surface
K = 0.00000e+00	A 4 = -2.25980e-05	A 6 = 2.43830e-08	A 8 = -3.13250e-11	25th Surface
K = 0.00000e+00	A 4 = -2.22797e-05	A 6 = 2.16423e-08	A 8 = -1.32136e-11	Focal Length
84.00	FNO	1.44	Half Angle of View (°)	14.44
Image Height	21.64	Overall lens length	117.50	BF
16.31	Infinity Short Distance (-700 mm)	d 7	4.26	10.25
d11	13.31	7.33	d15	5.79
1.00	d20	1.00	5.79	d25
16.31	6.01	Lens Unit Data		
Lens Unit Starting Surface				
Focal Length	1	1	93.82	2
8	-111.97	3	12	405.35
4	16	46.82	5	21
-54.20				

Numerical Example 2

(69) TABLE-US-00002 UNIT: mm Surface Data Surface No. r d nd vd 1 75.910 7.60 1.76385
48.5 2 278.185 0.20 3 81.251 4.11 1.59282 68.6 4 137.891 0.83 5 75.102 13.89 1.49700 81.5
6 -89.633 1.86 1.89190 37.1 7 128.606 (Variable) 8 -1506.651 4.06 1.94594 18.0 9 -117.758
0.20 10 -19994.632 1.40 1.80610 33.3 11 52.584 (Variable) 12(SP) ∞ 4.81 13 -83.985 1.40
1.69895 30.1 14 47.790 10.25 1.87070 40.7 15 -110.778 (Variable) 16 41.915 8.74 1.69680 55.5
17 -173.265 1.40 1.69895 30.1 18 89.247 1.05 19* 88.271 8.63 1.85135 40.1 20* -134.588
(Variable) 21 71.964 4.69 1.92286 20.9 22 -316.497 1.20 1.77047 29.7 23 27.700 8.93 24*
-37.015 2.00 1.88202 37.2 25* -58.504 15.69 Image Plane ∞ Aspheric Data 19th Surface K =
0.00000e+00 A 4 = -4.61083e-06 A 6 = -2.59781e-09 A 8 = -1.59591e-12 20th Surface K =
0.00000e+00 A 4 = -1.47775e-06 A 6 = -2.06115e-09 A 8 = -1.05938e-12 24th Surface K =
0.00000e+00 A 4 = 1.57806e-05 A 6 = -2.19149e-08 A 8 = 4.05207e-11 25th Surface K =
0.00000e+00 A 4 = 1.37873e-05 A 6 = -1.94341e-08 A 8 = 4.25441e-11 Focal Length 83.00 FNO
1.25 Half Angle of View (°) 14.61 Image Height 21.64 Overall lens length 133.01 BF 15.69
Infinity Short Distance (-700 mm) d 7 6.52 16.70 d11 17.27 7.09 d15 5.29 1.00 d20 1.00 5.29 d25
15.69 5.18 Lens Unit Data Lens Unit Starting Surface Focal Length 1 1 125.21 2 8 -127.16 3 12
293.03 4 16 43.02 5 21 -45.00

Numerical Example 3

(70) TABLE-US-00003 UNIT: mm Surface Data Surface No. r d nd vd 1 86.709 7.20 1.61800
63.4 2 369.488 0.20 3 69.023 7.29 1.49700 81.5 4 197.730 0.20 5 61.193 14.21 1.49700 81.5
6 -127.175 2.00 1.87070 40.7 7 93.340 0.21 8 95.976 5.85 1.49700 81.5 9 178.462 (Variable)
10 392.711 3.45 1.94594 18.0 11 -223.486 0.20 12 247.577 1.85 1.83481 42.7 13 40.471
(Variable) 14(SP) ∞ 4.82 15 -63.837 1.40 1.72825 28.5 16 44.300 9.51 1.87070 40.7 17
-74.293 (Variable) 18 43.005 4.33 1.72916 54.7 19 141.112 1.40 1.69895 30.1 20 66.919 2.02
21* 114.498 8.30 1.76450 49.1 22* -112.430 (Variable) 23 89.498 2.96 1.92286 20.9 24
1396.785 1.20 1.67300 38.3 25 29.039 8.92 26* -57.271 2.00 1.58313 59.4 27* -140.719
17.24 Image Plane ∞ Aspheric Data 21st Surface K = 0.00000e+00 A 4 = -6.61737e-06 A 6 =
-5.27557e-09 A 8 = -7.47702e-12 22nd Surface K = 0.00000e+00 A 4 = -4.30497e-06 A 6 =
-4.81815e-09 A 8 = -3.46829e-12 26th Surface K = 0.00000e+00 A 4 = -1.91464e-05 A 6 =
2.29584e-08 A 8 = -2.64435e-11 27th Surface K = 0.00000e+00 A 4 = -1.95897e-05 A 6 =
2.30203e-08 A 8 = -1.63145e-11 Focal Length 99.00 FNO 1.44 Half Angle of View (°) 12.33
Image Height 21.64 Overall lens length 133.50 BF 17.24 Infinity Short Distance (-700 mm) d 9
3.01 11.26 d13 15.82 7.57 d17 6.91 1.00 d22 1.00 6.91 d27 17.24 3.35 Lens Unit Data Lens Unit
Starting Surface Focal Length 1 1 95.26 2 10 -96.82 3 14 240.85 4 18 52.62 5 23 -52.76

Numerical Example 4

(71) TABLE-US-00004 UNIT: mm Surface Data Surface No. r d nd vd 1 174.540 5.00 1.59349
67.0 2 289.447 0.20 3 98.775 13.84 1.43387 95.1 4 1295.204 0.20 5 84.449 9.87 1.49700
81.5 6 232.436 0.20 7 73.986 17.37 1.49700 81.5 8 -295.998 2.00 1.87070 40.7 9 69.370
4.61 10 70.644 6.30 1.49700 81.5 11 177.413 (Variable) 12 478.926 4.01 1.94594 18.0 13
-478.755 0.20 14 148.546 3.00 1.80400 46.5 15 46.627 (Variable) 16(SP) ∞ 4.56 17 -79.969
1.40 1.77250 49.6 18 800.000 6.02 1.51742 52.4 19 -53.956 (Variable) 20 41.447 4.25 1.83400
37.2 21 225.784 1.40 1.84666 23.8 22 51.524 3.72 23* -230.448 4.98 1.76450 49.1 24* -69.264
(Variable) 25 -106.542 1.49 1.84666 23.8 26 -80.086 1.20 1.49700 81.5 27 39.190 7.26 28 55.080
3.12 1.58313 59.4 29* 88.761 38.05 Image Plane ∞ Aspheric Data 23rd Surface K =
0.00000e+00 A 4 = -9.23397e-06 A 6 = -6.40417e-09 A 8 = -9.52243e-12 24th Surface K =
0.00000e+00 A 4 = -7.11589e-06 A 6 = -4.70098e-09 A 8 = -6.63629e-12 29th Surface K =
0.00000e+00 A 4 = -6.74980e-07 A 6 = -3.45420e-10 A 8 = -7.82351e-13 Focal Length 198.00
FNO 2.05 Half Angle of View (°) 6.24 Image Height 21.64 Overall lens length 188.50 BF 38.05
Infinity Short Distance (-700 mm) d11 4.82 11.16 d15 24.36 18.02 d19 14.09 1.00 d24 1.00 14.09
d29 38.05 10.80 Lens Unit Data Lens Unit Starting Surface Focal Length 1 1 119.40 2 12 -132.23
3 16 16396.06 4 20 84.65 5 25 -84.19

Numerical Example 5

(72) TABLE-US-00005 UNIT: mm Surface Data Surface No. r d nd vd 1 90.525 4.22 1.76385 48.5 2 209.644 0.20 3 61.314 6.71 1.59522 67.7 4 204.455 0.20 5 50.907 12.20 1.49700 81.5 6 -130.720 1.60 1.80610 33.3 7 72.226 (Variable) 8 155.728 3.81 1.94594 18.0 9 -273.024 0.20 10 493.162 1.40 1.67300 38.3 11 34.600 (Variable) 12(SP) ∞ 1.57 13 323.382 1.40 1.73465 28.3 14 24.447 6.95 1.87070 40.7 15 86.688 (Variable) 16 37.184 4.58 1.90043 37.4 17 282.676 1.40 1.68167 31.4 18 38.658 3.09 19* 86.149 6.00 1.85135 40.1 20* -89.241 (Variable) 21 355.236 5.36 2.00069 25.5 22 -41.762 1.20 1.77047 29.7 23 32.530 9.05 24 -28.121 1.20 1.85478 24.8 25 -48.897 0.15 26 207.303 4.46 2.00100 29.1 27 -93.322 13.00 Image Plane ∞ Aspheric Data 19th Surface K = 0.00000e+00 A 4 = -1.35818e-05 A 6 = -1.76204e-08 A 8 = -5.55965e-11 20th Surface K = 0.00000e+00 A 4 = -9.59369e-06 A 6 = -1.32570e-08 A 8 = -3.11565e-11 Focal Length 84.00 FNO 1.44 Half Angle of View ($^{\circ}$) 14.44 Image Height 21.64 Overall lens length 114.50 BF 13.00 Infinity Short Distance (-700 mm) d 7 3.99 10.11 d11 13.37 7.24 d15 6.20 1.00 d20 1.00 6.20 d27 13.00 2.09 Zoom Lens Unit Data Lens Unit Starting Surface Focal Length 1 1 93.36 2 8 -122.52 3 12 -524.52 4 16 41.31 5 21 -95.79

(73) Table 1 below summarizes various values in each numerical example.

(74) TABLE-US-00006 TABLE 1 Inequality Ex. 1 Ex. 2 Ex. 3 Ex. 4 Ex. 5 (1) 0.194 0.189 0.174 0.192 0.155 (2) 0.207 0.283 0.411 0.012 0.160 (3) -1.550 -1.845 -1.876 -2.352 -0.877 (4) -0.750 -0.653 -1.023 -1.497 -0.686 (5) 0.895 0.663 1.039 1.658 0.900 (6) 1.794 1.930 1.881 2.339 2.033 (7) 1.399 1.602 1.348 0.952 1.363 (8) 2.180 2.864 2.395 1.847 1.909 (9) 0.359 0.265 0.423 0.543 0.298 (10) 0.071 0.123 0.083 0.032 0.073 (11) -0.057 -0.052 -0.060 -0.066 -0.062

Image Pickup Apparatus

(75) Referring now to FIG. 11, a description will now be given of an example of a digital still camera (image pickup apparatus) using the optical system according to each example. In FIG. 11, reference numeral 10 denotes a camera body, and reference numeral 11 denotes one of the optical systems described in Examples 1 to 5. Reference numeral 12 denotes a solid-state image sensor (photoelectric conversion element) such as a CCD sensor or CMOS sensor, which is built in the camera body 10 and receives and photoelectrically converts an optical image formed by the optical system 11. The camera body 10 may be a so-called single-lens reflex camera having a quick turn mirror, or a so-called mirrorless camera without a quick turn mirror.

(76) Applying the optical system according to each example to an image pickup apparatus such as a digital still camera can provide an image pickup apparatus having a small lens.

(77) Each of the above examples can provide an optical system having a long focal length, a large aperture ratio, a small size, high optical performance, and high-speed AF, and an image pickup apparatus having the optical system.

(78) While the disclosure has been described with reference to exemplary embodiments, it is to be understood that the present disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(79) This application claims the benefit of Japanese Patent Application No. 2022-000014, filed on Jan. 1, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. An optical system comprising, in order from an object side to an image side, a first lens unit having positive refractive power, a second lens unit having negative refractive power, a third lens unit having positive or negative refractive power, a fourth lens unit having positive refractive power, and a fifth lens unit having negative refractive power, wherein during focusing from infinity to the shortest distance, the second lens unit and the fourth lens unit move, the first lens unit, the third lens unit, and the fifth lens unit do not move, and wherein the following inequalities are

satisfied:

$$0.10 < sk/f < 0.25$$

$$0.010 < f/|f_3| < 0.500$$

$$-3.50 < f/f_5 < -0.80$$

$-1.6 < f/f_2 < -0.3$ where sk is a distance on an optical axis from a lens surface closest to an image plane of the optical system to the image plane, f is a focal length of the optical system in an in-focus state at infinity, and f_3 is a focal length of the third lens unit, f_5 is a focal length of the fifth lens unit, and f_2 is a focal length of the second lens unit.

2. The optical system according to claim 1, wherein the following inequality is satisfied:

$$0.4 < f/f_1 < 2.5$$
 where f_1 is a focal length of the first lens unit.

3. The optical system according to claim 1, wherein the following inequality is satisfied:

$$1.0 < f/f_4 < 3.5$$
 where f_4 is a focal length of the fourth lens unit.

4. The optical system according to claim 1, wherein the following inequality is satisfied:

$$0.7 < L/f < 2.0$$
 where L is a distance on the optical axis from a lens surface closest to an object of the optical system to the image plane.

5. The optical system according to claim 1, wherein the following inequality is satisfied:

$$1.4 < b_2 < 4.0$$
 where b_2 is a lateral magnification of the second lens unit in the in-focus state at infinity.

6. The optical system according to claim 1, wherein the following inequality is satisfied:

$$0.1 < b_4 < 0.8$$
 where b_4 is a lateral magnification of the fourth lens unit in the in-focus state at infinity.

7. The optical system according to claim 1, wherein the following inequality is satisfied:

$$0.01 < M_2/f < 0.30$$
 where a direction moving toward the image side is set positive and M_2 is a moving amount of the second lens unit during focusing from infinity to the shortest distance.

8. The optical system according to claim 1, wherein the following inequality is satisfied:

$$-0.20 < M_4/f < -0.01$$
 where a direction moving toward the image side is set positive, and M_4 is a moving amount of the fourth lens unit during focusing from infinity to the shortest distance.

9. The optical system according to claim 1, wherein the second lens unit includes, in order from the object side to the image side, a positive lens **L21** and a negative lens **L22**.

10. The optical system according to claim 9, wherein a biconcave air lens is formed between the positive lens and the negative lens.

11. The optical system according to claim 1, wherein the fourth lens unit includes, in order from the object side to the image side, a positive lens **L41**, a negative lens **L42**, and a positive lens **L43**, and wherein the positive lens **L41** and the negative lens **L42** are formed as a cemented lens.

12. The optical system according to claim 1, wherein the third lens unit includes, in order from the object side to the image side, a negative lens **L31** and a positive lens **L32**, and wherein the negative lens **L31** and the positive lens **L32** are formed as a cemented lens.

13. The optical system according to claim 1, wherein the third lens unit includes an aperture stop.

14. The optical system according to claim 1, wherein the first lens unit includes a positive lens **L11** disposed closest to an object.

15. The optical system according to claim 1, wherein the first lens unit includes two or more positive lenses and one or more negative lenses.

16. The optical system according to claim 1, wherein the fifth lens unit includes a positive lens and a negative lens.

17. The image pickup apparatus comprising: an optical system; and an image sensor configured to receive an image formed by the optical system, wherein the optical system includes, in order from an object side to an image side, a first lens unit having positive refractive power, a second lens unit having negative refractive power, a third lens unit having positive or negative refractive power, a fourth lens unit having positive refractive power, and a fifth lens unit having negative refractive power, wherein during focusing from infinity to the shortest distance, the second lens unit and the

fourth lens unit move, the first lens unit, the third lens unit, and the fifth lens unit do not move, and wherein the following inequalities are satisfied:

$$0.10 < sk/f < 0.25$$

$$0.010 < f/|f_3| < 0.500$$

$$-3.50 < f/f_5 < -0.80$$

$-1.6 < f/f_2 < -0.3$ where sk is a distance on an optical axis from a lens surface closest to an image plane of the optical system to the image plane, f is a focal length of the optical system in an in-focus state at infinity, and f_3 is a focal length of the third lens unit, f_5 is a focal length of the fifth lens unit, and f_2 is a focal length of the second lens unit.
